Shunchang Liu

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RESEARCH INTERESTS

My research focuses on **AI Alignment**. I'm currently most interested in the underlying principles of alignment, that is, understanding why models exhibit misaligned behaviors that deviate from human expectations, and how to control them from first principles. I've also worked on *AI safety* and *copyright*.

EDUCATION

EPFL, Switzerland — Sep 2026 — Present
PhD of Computer Science

EPFL - ETH Zurich (joint-degree program), Switzerland — Sep 2024 — Aug 2026
Master of Computer Science: Cybersecurity — GPA: 5.55 / 6.00 (Distinction)

Beihang University, China — Sep 2017 — Jun 2021
Bachelor of Engineering: Automation — GPA: 3.87 / 4.00 (Distinction)

WORK EXPERIENCE

Theory of Machine Learning (TML) Lab — Lausanne, Switzerland
Doctoral Researcher, Advisor: Prof. Nicolas Flammarion and Prof. Francesco Croce — Sep 2026 — present

Concordia AI (Social Enterprise focused on AI Safety and Governance) — Beijing, China
Intern, Mentor: Brian Tse — Apr 2024 — Aug 2024, Jun 2025 — Aug 2025

Beihang University — Beijing, China
Research Assistant, Advisor: Prof. Xianglong Liu — Sep 2021 — Dec 2023

SELECTED RESEARCH EXPERIENCE

Emergent Misalignment in Vision-Language Models — ETH Zurich
Master Thesis, Supervisor: Prof. Florian Tramèr and Prof. Francesco Croce — Feb 2026 — Aug 2026

- Investigating emergent misalignment in vision-language models (VLMs), where training on narrowly misaligned visual data induces broadly misaligned unsafe behaviors across unrelated domains and modalities.

Preference Instability Detection and Mitigation in Reward Models [paper] — ETH Zurich
Semester Project, Supervisor: Prof. Andreas Krause and Dr. Xin Chen — Mar 2025 — Dec 2025

- Identified preference instability in RLHF reward models, where semantic-preserving perturbations elicit contradictory preferences, and attributed it to unstable features isolated via Sparse Autoencoders (SAEs).
- Proposed two mitigation strategies, SAE Feature Steering and SAE Residual Correction, that substantially reduce incorrect preferences on harmlessness and hallucination benchmarks without retraining the reward model.

Copyright Infringement Detection and Mitigation for Diffusion Models [paper] — EPFL
Semester Project, Supervisor: Prof. Boi Faltings and Dr. Zhuan Shi — Sep 2024 — Jan 2025

- Proposed CopyJudge, an automated framework that uses large vision-language models to detect copyright infringement by comparing AI-generated and copyrighted images using abstraction-filtration-comparison and multi-model debate, and reducing risk through prompt tuning and non-infringing noise exploration in latent space.

Adversarial Patch Generation for Crowd Counting [paper] — Beihang University
Research Assistant, Supervisor: Prof. Aishan Liu and Dr. Yingwei Li (JHU) — Jun 2021 — Aug 2022

- Proposed a novel perceptual adversarial patch generation framework that exploited model-inherent perceptual properties, e.g., scale and position perceptions, of crowd counting models, which led to the SOTA attack.

- Employed adversarial training with our patches to improve models' cross-dataset generalization and robustness towards complex backgrounds, which showcased evidence of its beneficial impact on vanilla models' performance.

PUBLICATIONS

Co-first author (*) / Corresponding author (†)

Preference Instability in Reward Models: Detection and Mitigation via Sparse Autoencoders.
Shunchang Liu, Xin Chen, Belén Martín-Urcelay, Francesco Croce.
Preprint, 2026 [paper] [code]

CopyJudge: Automated Copyright Infringement Identification and Mitigation in Text-to-Image Diffusion Models.
Shunchang Liu*, Zhuan Shi*, Lingjuan Lyu, Yaochu Jin, Boi Faltings.
ACM Multimedia, 2025 [paper] [code]

Harnessing Perceptual Adversarial Patches for Crowd Counting.
Shunchang Liu*, Jiakai Wang*, Aishan Liu, Yingwei Li, Yijie Gao, Xianglong Liu, Dacheng Tao.
ACM CCS, 2022 [paper] [code]

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IP Protection in the Era of Visual Generative AI: A Survey.
Zhuan Shi, **Shunchang Liu**, Alireza Dehghanpour Farashah, Qian Yang, Han Yu, Cao Yang, Chaochao Chen, Yuping Yan, Yaochu Jin, Golnoosh Farnadi, Lingjuan Lyu.
Preprint, 2026 [paper]

The Biased Oracle: Assessing LLMs' Understandability and Empathy in Medical Diagnoses.
Jianzhou Yao*, **Shunchang Liu***, Guillaume Drui, Rikard Pettersson, Alessandro Blasimme, Sara Kijewski.
NeurIPS Workshop @GenAI4Health, 2025 [paper] [code]

Boosting Cross-task Transferability of Adversarial Patches with Visual Relations.
Tony Ma, Songze Li, Yisong Xiao, **Shunchang Liu†**.
IEEE CVPR Workshop @AdvCV, 2023 [paper]

Benchmarking the Robustness of Quantized Models.
Yisong Xiao, Tianyuan Zhang, **Shunchang Liu**, Haotong Qin.
IEEE CVPR Workshop @AdvCV, 2023 [paper]

Revisiting Audio Visual Scene-Aware Dialog.
Aishan Liu, Huiyuan Xie, Xianglong Liu, Zixin Yin, **Shunchang Liu**.
Neurocomputing, 2022 [paper]

Dual Attention Suppression Attack: Generate Adversarial Camouflage in Physical World.
Jiakai Wang, Aishan Liu, Zixin Yin, **Shunchang Liu**, Shiyu Tang, Xianglong Liu.
IEEE CVPR, 2021 (**Oral**) [paper] [code]

AWARDS

- EPFL EDIC PhD Fellowship — May 2026
- Outstanding Graduate of Beihang University **(Top 20%)** — Jun 2021
- CVPR Security AI Challenge: No-limit Adversarial Attacks on ImageNet **(11/1559)** — Apr 2021
- Outstanding Student of Beihang University **(Top 5%)** — Jun 2019

SERVICES

- **Program Committee / Reviewer:** NeurIPS [2026], Pattern Recognition, Frontiers of Computer Science, IEEE TCSVT, IEEE T-ITS, and Workshops (ICML MoFA 2025, CVPR AdvCV 2023, etc.)
- **Teaching Assistant:** Machine Learning (EPFL Fall 2025)